Lecture 04 — Use Case Modeling

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At the end of this session students should be able to:

1. **Define use case modeling.**
2. Distinguish **functional vs non-functional requirements**.
3. Define **actor, scenario, use case, and system boundary**.
4. Identify **primary, supporting, and offstage actors**.
5. Identify **actors and their goals** from a given scenario.
6. Convert **user goals into use cases**.
7. Give a suitable **use case name using verb + object**.
8. Distinguish **main success scenario and extensions**.
9. Explain **Brief, Casual, and Fully Dressed** formats.
10. Construct a **fully dressed use case** from a given scenario.
11. Identify **Summary, User Goal, and Sub-function** levels.

1. What is Use Case Modeling?

Use Case Modeling is a **requirements engineering technique** used to identify, elicit, and document the **functional requirements** of a system.

It describes **how users interact with a system to achieve their goals**.

- Focuses on **user goals and interactions**
- Describes the system from the **user's perspective**

- Use cases are primarily **written descriptions**, not diagrams.
- A **Use Case Diagram** in UML visually shows actors, use cases, and their relationships.

Functional Requirement

A statement of **what the system should do**.

Example:

- Validate ATM card.
- Validate PIN.
- Dispense cash.

Non-Functional Requirement

A **constraint on how the system should operate or be implemented**.

Example:

- Validate PIN within 3 seconds.
- Use encryption.
- Use a specific programming language.

Let’s go through case study sheet 1

1. What is a Use Case?

A **use case is a story describing how an actor uses a system to achieve a goal**.

Example: ATM

Actor: Customer
Goal: Withdraw money

The customer:

1. Inserts ATM card.
2. Enters PIN.
3. Selects "Withdraw Cash".
4. Enters amount.
5. System checks the account.
6. System dispenses cash.

This whole interaction represents the use case

Withdraw Cash

A use case is actually a **collection of different scenarios**, including successful and unsuccessful situations.

2. Use Case vs Scenario

Use Case

A **general goal/process**.

Scenario

One **specific path** through that use case.

Example

Use Case: Withdraw Cash

Possible scenarios:

Scenario 1 – Successful withdrawal

Customer enters correct PIN and withdraws Rs. 10,000.

Scenario 2 – Wrong PIN

Customer enters an incorrect PIN.

Scenario 3 – Insufficient balance

Customer requests Rs. 50,000 but has only Rs. 20,000.

Scenario 4 – Daily limit exceeded

Customer has already withdrawn the maximum allowed amount.

So:

Use Case = collection of related scenarios

3. Functional Requirement vs Use Case

Functional Requirement

Describes **what the system should do**.

Example:

The ATM shall validate the customer's PIN.

The ATM shall dispense cash.

The ATM shall display the account balance.

Use Case

Describes **how an actor interacts with the system to achieve a goal**.

Example:

Use Case: Withdraw Cash

Customer → Insert card

Customer → Enter PIN

System → Validate PIN

Customer → Enter amount

System → Check balance

System → Dispense cash

Functional requirement = What the system does

Use case = How the actor uses the system to achieve a goal

4. Non-Functional Requirement

A non-functional requirement describes a **constraint or quality requirement**.

Example for ATM:

Functional

The ATM shall validate the PIN.

Non-functional

The ATM shall validate the PIN within 3 seconds.

Other examples:

- Communication must use encryption.
- System should be available 99.9% of the time.
- Passwords must be securely stored.
- The system should respond within 2 seconds.

5. Actor

An **actor interacts with the system to achieve a goal**.

An actor can be:

- Person
- Organization
- Another computer system

The important point is that an actor is identified by its **role**, not necessarily its name.

Example: University Library System

Possible actors:

- Student
- Librarian
- Payment System
- University Authentication System

Not:

"Kamal"

Instead:

Student

because many different students can perform the same role.

6. Primary Actor

The **primary actor has a goal that is fulfilled by using the system**.

Example: Online Shopping System

Primary actor: Customer

Goal:

Purchase Product

The customer directly uses the system to achieve the goal.

Another example:

System	Primary Actor	Goal
ATM	Customer	Withdraw Cash
Library	Student	Borrow Book
Hospital	Patient	Book Appointment
Online Shopping	Customer	Place Order
University System	Student	Register Course

7. Supporting Actor

A **supporting actor provides a service to the system**.

Example: Online Shopping

Customer places an order.

The shopping system needs payment authorization.

So:

Primary Actor: Customer

Supporting Actor: Bank/Payment Gateway

Customer

↓

Online Shopping System

↓

Payment Gateway

The payment gateway is not using the shopping system to achieve its own user goal. Instead, it **provides a service to the system**.

8. Offstage Actor

This is an actor who **has an interest in the system's behavior but does not directly participate in the interaction**.

Example: Online Shopping

Suppose the shopping system calculates and reports sales tax.

The:

Customer → Primary actor

Payment Gateway → Supporting actor

Tax Authority → Offstage actor

The tax authority may care whether the correct tax is calculated, even though it doesn't directly interact with the system during the purchase.

9. One Example Showing All Actor Types

Consider a **Restaurant Ordering System**.

Scenario

A customer orders food using a restaurant POS system.

Actor	Type	Why?
Customer	Primary	Wants to place an order
Cashier	Primary	Processes the order
Payment Gateway	Supporting	Authorizes payment
Kitchen System	Supporting	Receives order information
Tax Authority	Offstage	Has an interest in tax calculation

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Case Study 1 – Online University Course Registration System

A university introduces an **Online Course Registration System**. Students use the system to view available courses, register for courses, drop courses, and view their registered courses.

The **Academic Staff** use the system to manage course information and approve special course-registration requests.

The system connects with the **University Payment System** to verify payments for courses that require fees. It also connects with the **University Student Information System** to retrieve student information and academic records.

The **University Academic Administration** is interested in ensuring that course registration follows university rules and that accurate registration records are maintained, but does not directly use the system.

Identify:

1. **Primary Actors** – At least 2
2. **Supporting Actors** – At least 2
3. **Offstage Actors** – At least 2

Answer

Actor	Type	Reason
Student	**Primary Actor**	Uses the system to register/drop courses and view courses
Academic Staff	**Primary Actor**	Manages courses and handles registration requests
University Payment System	**Supporting Actor**	Provides payment verification services
Student Information System	**Supporting Actor**	Provides student and academic information
Academic Administration	**Offstage Actor**	Has an interest in correct registration and university rules

University Management	**Offstage Actor**	Has an interest in registration reports and academic planning

Case Study 2 – Hospital Appointment and Prescription System

A **Hospital Appointment and Prescription System** allows patients to search for doctors, book appointments, cancel appointments, and view their appointment history.

Doctors use the system to view their appointments and issue electronic prescriptions.

The system communicates with the **Hospital Payment System** to process appointment payments and with an **SMS Notification Service** to send appointment reminders to patients.

The **Hospital Administration** is interested in ensuring that appointments are properly managed and that hospital policies are followed. The **Ministry of Health** is also interested in ensuring that prescription and patient-service activities comply with healthcare regulations, but neither directly uses the system.

Identify:

1. **Primary Actors** – At least 2
2. **Supporting Actors** – At least 2
3. **Offstage Actors** – At least 2

Answer

Actor	**Type**	**Reason**
Patient	**Primary Actor**	Books/cancels appointments and views appointment information
Doctor	**Primary Actor**	Manages appointments and issues prescriptions
Hospital Payment System	**Supporting Actor**	Processes appointment payments

SMS Notification Service	**Supporting Actor**	Sends appointment reminders
Hospital Administration	**Offstage Actor**	Has an interest in proper appointment management and policies
Ministry of Health	**Offstage Actor**	Has an interest in regulatory compliance

10. System Boundary

The **system boundary defines what is inside the system and what is outside**.

Example: Online Shopping System

Inside boundary:

Online Shopping System

- Search Products
- Place Order
- Make Payment
- Track Order

Outside:

- Customer
- Payment Gateway
- Delivery Company

The **actor is outside the system boundary**.

The **use cases are inside the system boundary**.

11. Actor-Goal List

After identifying actors, identify what each actor wants to achieve.

Example: **University Library System**

Actor	**Goals**
Student	Search Books
Student	Borrow Books
Student	Return Books
Student	View Borrowed Books
Librarian	Add Book
Librarian	Remove Book
Librarian	Update Book
Payment System	Process Fine Payment

It is recommend recording primary actors and their user goals in an **actor-goal list**.

12. How to Identify Actors

Ask yoursef:

Question 1

Who directly uses the system?

Example:

Student

Question 2

Who has different roles or permissions?

Example:

Student, Librarian, Administrator

Question 3

Who interacts with the system during errors?

Example:

Customer Service Officer

Question 4

Who provides a service to the system?

Example:

Payment Gateway

Question 5

Who is affected by the system's results?

Example:

Tax Authority

These questions are directly aligned with the actor-identification approach.

13. One Use Case for Each User Goal

The basic rule is:

One user goal → one use case

Example:

Student goals:

- Search Book
- Borrow Book
- Return Book
- View Borrowed Books

Therefore:

Student

|

+---- Search Book

|

+---- Borrow Book

|

+---- Return Book

|

+---- View Borrowed Books

It is recommend defining one use case for each user goal.

14. Exception: CRUD

Sometimes several goals can be combined.

CRUD means:

- Create
- Read
- Update
- Delete

Instead of:

- Create User
- View User
- Update User
- Delete User

we can use:

Manage Users

15. Naming a Use Case

Use Verb + Object

Examples:

✅ **Purchase Concert Ticket**

✅ **Borrow Book**

✅ **Search Book**

✅ **Place Order**

✅ **Register Course**

Avoid:

❌ Ticket Purchase

❌ Ticket Order

❌ Purchase

❌ Pay

Why?

Because the use case should:

- Represent one clear goal
- Be from the actor's viewpoint
- Use active voice
- Usually use verb–object form

16. Main Success Scenario

This means:

What happens when everything goes correctly?

Example: Borrow Book

Main Success Scenario

1. Student searches for a book.
2. System displays available books.
3. Student selects a book.
4. System checks availability.
5. System records the borrowing.
6. System displays the due date.

That's the **normal successful path**.

17. Alternate / Failure Scenarios

What can go wrong?

For **Borrow Book**:

Extension 1 – Book unavailable

Student selects a book.

→ System finds that the book is unavailable.

→ System displays:

"Book currently unavailable."

Extension 2 – Student has unpaid fine

→ System checks student's account.

→ Unpaid fine exists.

→ System prevents borrowing.

Extension 3 – Student has reached borrowing limit

→ System informs the student that the borrowing limit has been reached.

18. Use Case Formats

There are three **formats**:

1. Brief

Very short summary.

2. Casual

Informal description containing several scenarios.

3. Fully Dressed

Detailed, structured description containing steps, variations and supporting information.

19. Brief Use Case Example

Use Case: Borrow Book

A student searches for a book, selects an available book and borrows it. The system records the borrowing and displays the due date.

Very short.

When useful?

For simple, well-understood functionality.

20. Casual Use Case Example

Use Case: Handle Return

A customer arrives at the checkout with items to return. The cashier records the returned items using the POS system. If the refund is approved, the system processes the refund. If the refund is rejected, the cashier uses another refund method. If the item cannot be identified, the system asks the cashier to enter the item code manually.

This is more detailed but still written as normal paragraphs.

21. Fully Dressed Use Case

This is the most important format.

A fully dressed use case gives structured details such as:

- Use Case Name
- Primary Actor
- Stakeholders and Interests
- Preconditions
- Main Success Scenario

- Extensions
- Postconditions / Success Guarantees

22. Fully Dressed Example – Online Library.

Use Case: Search Book

Primary Actor: Student

Precondition

- Student is logged in.
- Books are registered in the system.

Main Success Scenario

1. Student enters book name, author name or ISBN.
2. Student selects **Search**.
3. System validates the search details.
4. System searches for matching books.
5. System displays book details.

Extensions

3a. Invalid search details

1. System detects invalid parameters.
2. System displays:

"Please enter valid details."

4a. Book not found

1. System cannot find a matching book.
2. System displays:

"No search results to show."

23. Understanding Precondition

A **precondition** is something that must already be true **before the use case starts**.

Borrow Book

Preconditions:

- Student is logged in.
- Book exists in the library system.

Withdraw Cash

Preconditions:

- ATM is operational.
- Customer has inserted a valid card.

Place Order

Preconditions:

- Customer is logged in.
- Products are available.

24. Understanding Postcondition

A **postcondition** describes what should be true **after the use case finishes**.

Borrow Book

After successful completion:

- Book is marked as borrowed.
- Student's borrowed-book list is updated.
- Due date is recorded.

Place Order

After successful completion:

- Order is created.
- Payment is recorded.
- Product quantity is updated.
- Order confirmation is generated.

25. Stakeholders and Interests

A stakeholder is someone who has an interest in the outcome of the use case.

Example: Place Order

Stakeholder	**Interest**
Customer	Wants order placed successfully
Store	Wants correct order and payment
Payment provider	Wants valid payment
Delivery company	Needs correct delivery information
Tax authority	Wants correct tax calculation

26. Complete Example – Online Shopping

Let's put **everything together**.

System

Online Shopping System

System Boundary

Inside:

- Product search
- Cart
- Order placement
- Payment processing
- Order tracking

Outside:

- Customer
- Payment Gateway
- Delivery Company

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Actors

Primary

- Customer
- Admin

Supporting

- Payment Gateway
- Delivery Service

Offstage

- Tax Authority

Customer Goals

- Browse Products
- Place Order
- Track Order

Admin Goals

- Manage Products
- View Sales Reports

Use Cases

Customer

```
|
+---- Browse Products
|
+---- Place Order
|
+---- Track Order
```

Admin

```
|
+---- Manage Products
|
+---- View Sales Reports
```

27. How to Solve a Use Case

Step 1 — Find the system

What system are we studying?

Example:

University Library System

Step 2 — Find actors

Ask:

Who uses it?

Example:

- Student
- Librarian

Step 3 — Find goals

Ask:

What does each actor want to achieve?

Student:

- Search Book
- Borrow Book
- Return Book

Librarian:

- Add Book
- Remove Book
- Update Book

Step 4 — Convert goals into use cases

Use **Verb + Object**:

- Search Book
- Borrow Book
- Return Book
- Manage Books

Step 5 — Find scenarios

Ask:

What normally happens?

Step 6 — Find exceptions

Ask:

What can go wrong?

Step 7 — Choose the required format

If asked for:

Brief → short paragraph

Casual → informal description with scenarios

Fully dressed → structured template

29. Quick Comparison Table

Concept	Simple Meaning	Example
System	What we are studying	Library System
Actor	Who interacts with system	Student
Primary Actor	Has the goal	Student
Supporting Actor	Provides service	Payment Gateway
Offstage Actor	Has an interest	Tax Authority
Goal	What actor wants	Borrow Book
Use Case	Complete goal/process	Borrow Book
Scenario	One particular path	Successfully borrow book
Main Success	Normal successful flow	Book available → borrowed
Extension	Alternative/failure flow	Book unavailable
Precondition	Must be true before	Student logged in

Postcondition	Result after completion	Book marked as borrowed
System Boundary	Separates system from outside	Library system boundary
Brief	Very short description	One paragraph
Casual	Informal multiple scenarios	Short paragraphs
Fully Dressed	Detailed structured description	Template

Activities

For each question below , **write a fully dressed use case**

In Class

Question 1 – Library Book Search

An **Online Library System** allows students to search for books using the book title, author name, or ISBN. The student must first log in. If the login is successful, the student can enter search details. If valid details are entered and the book exists, the system displays the book details. If the details are invalid, the system displays an error message. If no matching book is found, the system displays a "No results found" message.

Write a fully dressed use case for: *Search Book*.

Question 2 – ATM Cash Withdrawal

An **ATM System** allows customers to withdraw cash. The customer inserts an ATM card and enters a PIN. If the PIN is correct, the customer enters the amount to withdraw. The system checks the account balance and daily withdrawal limit. If everything is valid, the ATM dispenses cash and updates the account balance.

If the PIN is incorrect, the system displays an error. After three incorrect attempts, the card is blocked. If the account has insufficient funds or the daily limit has been exceeded, the withdrawal is rejected.

Write a fully dressed use case for: *Withdraw Cash*.

Question 3 – Online Shopping

An **Online Shopping System** allows customers to purchase products. The customer searches for a product, selects a product, adds it to the cart, and proceeds to checkout. The customer enters delivery details and selects a payment method. If payment is successful, the system creates the order and displays an order confirmation.

If the product is out of stock, the system informs the customer. If payment fails, the customer is asked to select another payment method.

Write a fully dressed use case for: *Place Order*.

Question 4 – University Course Registration

A **University Course Registration System** allows students to register for courses. The student logs into the system and views available courses. The student selects a course and submits the registration request. The system checks whether the student satisfies the prerequisites and whether seats are available. If all conditions are satisfied, the student is registered for the course.

If the prerequisite is not satisfied, registration is rejected. If the course is full, the system informs the student that registration is unavailable.

Write a fully dressed use case for: *Register for Course*.

Question 5 – Hospital Appointment Booking

A **Hospital Appointment System** allows patients to book appointments with doctors. The patient logs into the system, selects a doctor and preferred date, and views available time slots. The patient selects a time slot and confirms the appointment. The system records the appointment and sends a confirmation.

If no time slots are available, the system informs the patient. If the selected slot becomes unavailable before confirmation, the system asks the patient to select another slot.

Write a fully dressed use case for: *Book Appointment*.

Question 6 – Online Banking Login

An **Online Banking System** allows registered users to securely access their accounts. The user enters a username and password. The system verifies the credentials. If they are correct, the user is redirected to the account dashboard.

If the password is incorrect, the system displays an error and allows another attempt. After three consecutive failed attempts, the account is locked. If the user forgets the password, the user can select **Forgot Password** and reset it using email or SMS verification.

Write a fully dressed use case for: *Login to Online Banking*.

Question 7 – Food Delivery Order

A **Food Delivery System** allows customers to order food from restaurants. The customer selects a restaurant, chooses food items, adds them to the cart, enters the delivery address, and confirms the order. The system processes the payment and sends the order to the restaurant.

If a selected food item becomes unavailable, the system asks the customer to remove or replace it. If payment fails, the order is not confirmed.

Write a fully dressed use case for: *Place Food Order*.

Question 8 – Hotel Room Booking

An **Online Hotel Booking System** allows customers to search for available rooms. The customer enters the check-in and check-out dates and selects a room. The system displays available rooms and their prices. The customer provides guest details and confirms the booking. After successful payment, the system confirms the reservation.

If no rooms are available, the system displays an appropriate message. If payment fails, the booking is not completed.

Write a fully dressed use case for: *Book Hotel Room*.

Question 9 – Prescription Refill

A **Hospital System** allows patients to request prescription refills online. The patient logs in and views active prescriptions. The patient selects a prescription and submits a refill request. The system checks whether the prescription is still valid and whether the refill limit has been exceeded. If eligible, the request is sent to the doctor for approval.

If the prescription is expired or the refill limit has been exceeded, the system rejects the request. If the doctor approves the request, the system updates the prescription status and notifies the patient.

Write a fully dressed use case for: *Request Prescription Refill*.

Question 10 – Cinema Ticket Booking

An **Online Cinema Booking System** allows customers to select a movie, date, show time, and seats. The customer confirms the selected seats and makes payment. If payment is successful, the system generates the ticket and sends a confirmation.

If the selected seats are already booked, the system asks the customer to select different seats. If payment fails, the booking is not completed.

Write a fully dressed use case for: *Book Cinema Ticket*.

Question 11 – University Examination Registration

A **University Examination System** allows students to register for upcoming examinations. The student logs in and views eligible examinations. The student selects examinations and submits the registration. The system checks whether the student has completed the required courses and whether registration is still open. If all conditions are satisfied, the registration is recorded and a confirmation is displayed.

If the student is not eligible for an examination, the system rejects the registration. If the registration deadline has passed, the system does not allow registration. If the student has already registered, the system informs the student.

Write a fully dressed use case for: *Register for Examination*.

Question 12 – Bank Fund Transfer

An **Online Banking System** allows customers to transfer money between accounts. The customer logs in, selects the source account, enters the recipient account and transfer amount, and confirms the transaction. The system verifies the customer's identity, checks the account balance, and processes the transfer. If successful, the system updates both accounts and displays a transaction confirmation.

If the balance is insufficient, the transfer is rejected. If the recipient account is invalid, the system displays an error. If additional authentication is required and fails, the transaction is cancelled.

Write a fully dressed use case for: *Transfer Money*.